Combinatories

Sequence:- A set of quantities written according to definite rule.

e.g:- Let us consider a sequence $S = \{3^1, 3^2, 3^3, \ldots, 3^n, \ldots\}$

$$S = \{S_n\}, \text{ where } S_n = 3^n$$

$$S_n = 3S_{n-1}, \quad n \geq 2, \quad S_1 = 3$$

Recurrence relation

Definition:- A recurrence relation for the sequence $\{S_n\}$ is an equation that relates S_n in terms of one or more of the previous terms of the sequence, namely $S_0, S_1, S_2, \ldots, S_{n-1}$ $\forall$ integer $n \geq n_0$, where n_0 is non negative integer.

Q. Show that the sequence

$$\{2, 3, 4, 5, \ldots, 2+n, \ldots\}$$

for $n \geq 0$ satisfies the recurrence relation (or difference equation) $a_k = 2a_{k-1} - a_{k-2}$, $k \geq 2$

Solution:- Let $a_n = 2+n$ (nth term of the sequence)

$$\therefore a_k = 2+k$$

$$a_{k-1} = 2+(k-1) = 1+k$$

$$a_{k-2} = 2+(k-2) = k$$

Now,

$$2a_{k-1} - a_{k-2}$$

$$= 2(1+k) - k$$

$$= 2+k$$

$$= a_k$$

$$\therefore a_k = 2a_{k-1} - a_{k-2}$$

2008

Q. Show that $y_k = \frac{1}{2}k(k-1)$ is the solution of the difference equation $y_{k+1} - y_k = k$.

Solution:- Given that

$$y_k = \frac{1}{2}k(k-1)$$

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Now,

$$y_{k+1} = \frac{1}{2}(k+1)(k+1-1) = \frac{1}{2}k(k+1)$$

Now,

$$y_{k+1} - y_k = \frac{1}{2}k(k+1) - \frac{1}{2}k(k-1)$$

$$= \frac{1}{2}k(k+1-k+1)$$

$$= k$$

$$\therefore y_{k+1} - y_k = k$$

2003

Q: Show that $y_k = 1 - \frac{2}{k}$, $k = 1, 2, 3, \ldots$ is a solution of the difference equation

$$(k+1)y_{k+1} + ky_k = 2k - 3$$

Solution:-

Given that,

$$y_k = 1 - \frac{2}{k}, \quad k = 1, 2, 3, \ldots \text{ is}$$

$$\therefore y_{k+1} = 1 - \frac{2}{(k+1)}$$

Now,

$$(k+1)y_{k+1} + ky_k$$

$$= (k+1)\left[1 - \frac{2}{k+1}\right] + k\left(1 - \frac{2}{k}\right)$$

$$= (k+1) - 2 + k - 2$$

$$= 2k - 3$$

$x = 3$ is a soln $x - 3 = 0$

$x - 3 = 3 - 3 = 0$

$x - 3 =$

Note:- Recurrence relation sometimes called difference equation.

Linear Recurrence relation with constant co-efficient

A linear recurrence relation with constant co-efficient is of the form

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} = f(n)$$

where c_i's are constants.

The order of the recurrence relation is the difference between the highest and lowest subscript of a.

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e.g:- ① The recurrence relation

$$a_n = a_{n-1} + a_{n-2}$$

$\therefore$ the order of this recurrence relⁿ is 2.

Homogeneous recurrence relation

A recurrence relation is of the form

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} = 0$$ is called homogeneous recurrence relation.

~~Non homogeneous recurrence relation~~

e.g:- The recurrence relation, $a_n = a_{n-1} + a_{n-2}$ is linear homogeneous ~~equation~~ relation.

Non homogeneous recurrence relation

A recurrence relation is of the form

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} = f(n)$$

is called non homogeneous relation.

e.g:- The recurrence relation

$$a_n - a_{n-1} = 3$$ is not a linear homogeneous relⁿ

Q. Form the difference equation of lowest order by eliminating the arbitrary constants a and b given by

$$y_k = a2^k + b3^k$$

or

2nd

Find the recurrence relation that is satisfied by the sequence

$$y_k = a2^k + b3^k$$

Solⁿ:- Given that

$$y_k = a2^k + b3^k \longrightarrow ①$$

$$y_{k-1} = a2^{k-1} + b3^{k-1}$$

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$$= \frac{a2^k}{2} + \frac{b3^k}{3}$$

$$= \frac{3a2^k + 2b3^k}{6}$$

$$\Rightarrow 6y_{k-1} = 3a2^k + 2b3^k$$

$$= 3(a2^k + b3^k) - b3^k$$

$$= 3y_k - b3^k$$

$$\Rightarrow b3^k = 3y_k - 6y_{k-1} \longrightarrow (ii)$$

Again,

$$y_{k-2} = a2^{k-2} + b3^{k-2}$$

$$= \frac{a2^k}{2^2} + \frac{b3^k}{3^2}$$

$$= \frac{a2^k}{4} + \frac{1}{9}[3y_k - 6y_{k-1}] \quad \text{using (ii)}$$

$$= \frac{1}{4}a2^k + \frac{1}{3}(y_k - 2y_{k-1})$$

$$= \frac{1}{4}[y_k - b3^k] + \frac{1}{3}(y_k - 2y_{k-1})$$

$$= \frac{1}{4}[y_k - 3y_k + 6y_{k-1}] + \frac{1}{3}(y_k - 2y_{k-1})$$

$$= \frac{1}{2}(3y_{k-1} - y_k) + \frac{1}{3}(y_k - 2y_{k-1})$$

$$\Rightarrow 6y_{k-2} = 9y_{k-1} - 3y_k + 2y_k - 4y_{k-1}$$

$$\Rightarrow y_k = 5y_{k-1} - 6y_{k-2}$$

which is the reqd. diff. eqn.

2002/9. Form the difference equation of lowest order by eliminating the arbitrary constants a and b given by

$$y_k = (a + bk)2^k.$$

Solution:- Given that,

$$y_k = (a + bk)2^k \longrightarrow (1)$$

$$\therefore y_{k-1} = (a + b(k-1))2^{k-1}$$

$$= \frac{(a + bk - b)2^k}{2}$$

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$$\Rightarrow 2y_{k-1} = (a+bk)2^k - b2^k$$

$$= y_k - b2^k \quad (\because \text{using } ①)$$

$$\Rightarrow b2^k = y_k - 2y_{k-1} \longrightarrow ②$$

Again, $y_{k-2} = [a+b(k-2)]\,2^{k-2}$

$$= \frac{(a+bk - 2b)\,2^k}{2^2}$$

$$= \frac{(a+bk)2^k - 2(b2^k)}{4}$$

$$\Rightarrow 4y_{k-2} = y_k - 2(y_k - 2y_{k-1})$$

$$\therefore y_k = 4(y_{k-1} - y_{k-2})$$

which is the reqd recurrence relation.

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